

BigQuery for Data Analysis

Mastering Marketing Data with BigQuery SQL & Studio

Luxottica Global Analytics & Digital Commerce Workshop

Project GCP: `qwiklabs-gcp-04-9efaa47f1d21`



Executive Mission Briefing: The Q4 Dilemma

- **The CMO's Challenge:** Global digital ad spend increased by +35%, but online e-commerce revenue growth remained flat at +2%.
- **The Core Question:** *Where is the marketing budget leaking, and how do we unlock high-ROAS growth for Luxottica?*
- **Your 3-Hour Mission:**
 1. Explore 116,000+ records in BigQuery Studio.
 2. Uncover budget waste on third-party networks.
 3. Prove the high-ROAS growth power of **Google Ads** (Search, Shopping, YouTube, PMax)

Gamification & Brand Detective Teams

4 Brand Detective Teams

-  **Team Ray-Ban** (Smart Glasses & Heritage Icons)
-  **Team Oakley** (Prizm Sport Technology)
-  **Team Persol** (Italian Handcrafted Luxury)
-  **Team Oliver Peoples** (High-AOV Luxury Eyewear)

Point Scoring System

- **+100 Points:** First team with the correct SQL query!
- **+50 Points:** Best business insight / storytelling interpretation.
- **+100 Points:** Most creative Looker Studio Executive Dashboard

BigQuery Studio: Data Canvas & Gemini AI

- **Visual Node Analytics (DAG):** Map relationships between `lux_crm_customers` → `lux_online_orders` → `lux_ad_spend`.
- **Natural Language Prompts:** Type plain Italian/English requests to auto-generate queries and charts.
- **Gemini Insights Node:** Automated executive summaries generated directly from data visualizations.

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-- Prompt in Data Canvas:  
-- "Show total sales and average order value by brand in dataset luxottica_marketing_analytics"
```

Chapter 1: High Margin Discovery

What the Data Reveals:

- **Oliver Peoples:** Top revenue leader (€3.96M) with premium AOV (€396.00).
 - **Ray-Ban Meta Smart Glasses:** High-margin category (€320.00+ AOV).
 - **Vogue Eyewear:** Lagging performance (€1.29M) due to heavy promotional discounting (€129.00 AOV).
- “ **Plot Twist #1:** Ray-Ban Meta Smart Glasses represent Luxottica's biggest margin expansion opportunity, but online sales are capped! Why? ”



Coffee Break (15 Minutes)

- Network with your Brand Detective Team!
- Review your score on the Leaderboard.
- Get ready for **Multi-Source Cross-Channel Analytics** in Chapter 2!

Chapter 2: The Google ROAS Revelation

Cross-Channel Performance Analysis (SQL JOINS)

Channel / Platform	Spend (€)	Revenue (€)	ROAS	Verdict
TikTok / Criteo (3rd-Party Social)	€2,200/day	Low	< 0.8x	✗ Budget Waste!
Google Search & Shopping	€450/day	Massive	6.8x - 8.2x	🚀 High-ROAS Hero!
YouTube Ads & PMax	€600/day	High	6.5x+	🚀 Scale Channel!

Chapter 3: Data Wrangling & Google Ads Customer Match

- **Dirty Data Problem:** 1,000 messy lead records in `lux_raw_marketing_leads_dirty` (untrimmed spaces, uppercase emails, mixed currency formats).
- **Gemini Visual Data Prep:** Clean data without writing manual string manipulation code!
- **350+ VIP Leads Recovered:** Leads with estimated spend > €200 siphoned directly into **Google Ads Customer Match** for high-converting retargeting!



The Q4 Google Growth Strategy

1. **Riallocate 60% of Budget:** Shift funds away from 0.7x ROAS social networks to **Google Search, Shopping, YouTube Ads, and Performance Max.**
2. **Capture 100% Impression Share:** Never run out of daily budget on Google Search for *Ray-Ban Meta Smart Glasses* and *Oliver Peoples*.
3. **Activate Customer Match:** Re-engage 350+ VIP leads on YouTube Ads and Google Search.
4. **1-Click Executive Dashboard:** Real-time growth tracking on Looker Studio!



Award Ceremony & Next Steps

Congratulations to the Winning Brand Detective Team!

- **Repository:** github.com/mauripsale/luxottica-bigquery-training
- **GCP Project:** [qwiklabs-gcp-04-9efaa47f1d21](https://console.cloud.google.com/projects/qwiklabs-gcp-04-9efaa47f1d21)

Thank you for mastering BigQuery Studio with Google Cloud!